

Abhiram Mokkaapati

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Education

Bachelor of Science in Electrical Engineering & Computer Science, Minor in Data Science

Expected May 2027

University of California, Berkeley

Relevant Coursework: Data Structures & Algorithms, Efficient Algorithms & Intractable Problems, Operating Systems, Computer Security, Computer Architecture, Artificial Intelligence, Principles & Techniques of Data Science, Discrete Math & Probability Theory

Skills

Programming: Python, Java, C++, C, Go, SQL, JavaScript, TypeScript

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, PyTorch, LightGBM, OpenCV, MediaPipe, YOLOv8, NumPy, Pandas

Tools & Platforms: Git, Docker, PostgreSQL, Supabase, Vercel, GitHub Actions, WebSockets

Work Experience

Motion Performance CoPilot, McQueen.AI

San Francisco, CA

Data Engineer → Engineering Lead

Sept 2025 - Jul 2026

- Led about five engineers and three interns as Engineering Lead of a real-time, multi-camera swim analytics platform, owning architecture, code review, hiring, and delivery.
- Shipped an end-to-end platform for the Cal swim team: GPU-accelerated multi-camera tracking (YOLOv8 + OSNet Re-ID), FastAPI inference services, Supabase, and a Next.js coach app supporting two cameras and wall-touch detection across 4-5 lanes.
- Built a boot-started on-deck server that opens a Cloudflare tunnel and registers its address in Supabase, so coaches control cameras and get the live WebSocket feed from any network.
- As Data Engineer, built real-time stroke classification (MediaPipe, OpenCV, temporal pattern analysis) across all four strokes, plus automated wall detection and lap tracking (direction-change algorithms, adaptive learning) for stroke counting, split times, and JSON session export.
- Designed a context-aware feedback engine comparing stroke counts and pace against intensity-based targets to generate personalized coaching insights.

Projects

Secure End-to-End Encrypted File Sharing System

Go, Cryptography – [Live demo](#)

- Designed and implemented from scratch a Dropbox-like client in Go with end-to-end encryption, user authentication, file operations, RSA-2048 signed sharing, and secure revocation via file re-encryption at randomized locations; wrote the full test suite and independently deployed it as a live web app.
- Engineered linked-list-based appends with a constant ~1,800-byte bandwidth overhead regardless of file size ($O(1)$ in file size), enabling multi-gigabyte file updates without re-downloading the file.

Hall Staff Duty – Multi-Tenant RA Scheduling Platform

Next.js, TypeScript, Supabase (PostgreSQL)

- Built for Unit 1 (six buildings) and used by every RA on staff; architected a multi-tenant platform enforcing Postgres Row-Level Security across 4 role tiers, verified via manual impersonation testing of each role to confirm no cross-tenant data leakage.
- Built a custom greedy and local-search swap algorithm to auto-generate fair on-call schedules under availability and consecutive-day constraints, and shipped invite-only auth with rotating-token .ics calendar feeds.

F1 Race Outcome Prediction Model

Python, LightGBM, FastAPI, React

- Redesigned race-outcome prediction from independent per-target classifiers into a learning-to-rank model (LightGBM, LambdaRank/NDCG) on historical FastF1 data; on the held-out 2023 season it achieved a 70% winner hit rate, 64% podium precision, and 0.66 average rank correlation.
- Validated a Plackett-Luce probability derivation that beat naive normalization on Brier score (0.0158 vs 0.0171 for wins), and used time-based train/test splits to prevent lookahead leakage into held-out seasons.

Leadership

- Senior Resident Assistant, UC Berkeley Residential Life: mentor and schedule a team of 10 RAs.
- Pool Manager, City of Concord: run operations and safety at a pool serving ~124,000 swimmers a year.